Disk Utilization with DAS

- If a server requires 100GB of storage for its OS and apps when deployed, and is expected to grow to 300GB of storage capacity, it's typical to install 500GB of disk space
- You want to leave some overhead for unexpected growth, and it might not be possible to add capacity later without bringing the server offline
- You'll often get utilization of around 30% when using Direct Attached Storage

Disk Utilization Example - DAS

- Initial requirement = 100GB per server
- 50 servers x 100GB = 5TB total
- Expected Size = 300GB per server
- 50 servers x 300GB = 15TB total
- What you buy: 50 x 500GB = 25TB total
- In use = 8TB total
- You pay for a lot of wasted space

Disk Utilization with Centralised Storage

- Storage systems provide a centralized ‘pool’ of shared storage
- Devices and applications can be allocated storage as required and on enterprise class storage the size can be easily changed on the fly, non-disruptively
- Centralised storage can provide disk utilization closer to 80%

Disk Utilization – Thin Provisioning

- Thin provisioning allows you to make it appear to the servers that they have more storage than you have actually paid for
- What you buy with centralized storage: 50 x 200GB = 10TB total
- The servers each believe they have 500GB space
- It looks like there is 25TB storage, but you only paid for 10TB

Disk Utilization – Thin Provisioning

- The 10TB shared space is used first-come first-served
- You can add the additional space as and when needed, transparently to the servers

Disk Utilization – Deduplication and Compression

- Deduplication detects and eliminates identical blocks
- The eliminated blocks are replaced with a pointer to a single copy of the block on disk
- Compression detects and eliminates redundant data and white space in files
- Huge space savings are possible, dependent on the amount of duplicate blocks and compressible files you have

Disk Utilization – Centralised Storage

- Pooled storage and thin provisioning moves from a ‘Just In Case’ to a ‘Just In Time’ model of purchasing storage space
- Deduplication and compression provide additional space savings
- This provides cost savings on hardware, rack space, power and cooling, and the savings are multiplied as storage costs tends to come down over time

Performance and Capacity

- Accessing data across a network rather than in DAS can add latency, but…
- Data can be striped across many disks in an enterprise class storage system
- Storage vendors are also at the cutting edge of new storage technologies

Resiliency

- SAN and NAS storage systems are always built to have very high degrees of resiliency because they will almost always be mission critical systems for the enterprise
- If any single component fails there is a redundant component to take its place

Benefits - Centralized Management

- It’s much easier to manage all your storage from a centralised location, rather than separately across all your servers

Diskless Servers

- SAN protocols allow clients to boot up from a logical disk on the remote storage
- The clients don’t need to contain any disks
- This is a very popular option with blade servers



Storage Tiering

- The storage system can have media with differing attributes, such as high performance SSD drives, and high capacity (but lower performance) SATA drives
- Frequently accessed ‘hot’ data can be kept on the SSDs, and ‘cold’ data automatically archived onto the SATA drives

Centralized Backup

- Managing backups is very inconvenient and time consuming when you have 50 different tape drives on 50 different servers
- Centralized storage provides a centralized backup solution
- Storage systems can also backup to remote disk (rather than tape) which reduces backup windows and doesn't require loading/unloading of physical media

Snapshots

- Snapshots are a point in time copy of the file system which can be used as a convenient short term backup
- They consist of pointers to the original blocks on disk rather than being a new copy of the data, so they initially take up no space and occur nearly instantaneously

Snapshots

- If data gets corrupted or somebody accidentally deletes a file then you can very quickly recover from a snapshot
- Snapshots do not replace a long term offsite backup solution, but they're great for short term very quick and convenient backups and restores

Disaster Recovery

- You can replicate data to a disaster recovery site
- You can also load balance incoming client requests for read-only data between the different sites. (You can't do this for writable data as you need to maintain one consistent copy of the data)

Virtualization Support - vMotion

- Software such as VMware and Hyper-V allows you to run multiple virtual servers on the same underlying physical hardware server.
- You can have a Linux web server, Exchange mail server and SQL database server for example all running on the same physical box, transparently to each of those virtual servers.

Virtualization Support - vMotion

- The killer feature of virtualisation software is the ability to move virtual servers between physical servers on the fly while they are still running.
- The virtual servers can keep on running with no outages even if the underlying physical server fails or is taken down for maintenance.
- External storage is a requirement for this feature.

